Lorenzo Dello Schiavo

Curriculum Vitæ

Contact Information

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Università degli Studi di Roma "Tor Vergata"
Via della Ricerca Scientifica 1
00133 Roma, Italy

Academic appointments

11/2024–present **Assistant Professor (Ricercatore in tenure track)**, *Università degli Studi di Roma "Tor Vergata"*, Rome, IT.

09/22–11/2024 **FWF ESPRIT Fellowship**, *Institute of Science and Technology Austria*, Klosterneuburg, AT.

09/20–06/22 **PostDoc**, *Institute of Science and Technology Austria*, Klosterneuburg, AT.

- Spezialforschungsbereich (Special Research Center) 65: *Taming Complexity in Partial Differential Systems*, Project P3: *Structure preserving variational discretisation via optimal transport*, PI: Prof. Dr. J. Maas

11/15–08/20 ***Scientific collaboration***, *IAM @ HCM*, Bonn, DE.

09/16–08/20 - Sonderforschungsbereich (Collaborative Research Center) 1060: *The mathematics of emergent effect*, Project B03: *Optimal transport and random measures*, PI's: Prof. Dr. K.-T. Sturm, Dr. M. Huesmann.

11/15–09/16 - Hausdorff Center for Mathematics.

Education

11/15–11/19 🎓 **Ph.D. in Mathematics (Dr. rer. nat.)**, *BIGS @ University of Bonn*, Bonn.

- **Thesis:** Diffusions on Wasserstein Spaces (supervisor: K.-T. Sturm)
- **Defense:** *viva* in English, November 26, 2019, *magna cum laude*

10/15 **Visiting studentship**, *IAM @ HCM*, Bonn, 🏛 Group of Stochastic Analysis.

10/13–09/15 🎓 **M.Sc. in Mathematics (*Laurea Magistrale*)**, *Sapienza – Università di Roma*, Rome.

- **Thesis:** *Heat equation on metric measure spaces* (in English, supervisor: A. Pisante)
- **Defense:** *viva* in Italian, September 21, 2015, with honors (*con lode*)

10/10–10/13 🎓 **B.Sc. in Mathematics (*Laurea Triennale*)**, *Sapienza – Università di Roma*, Rome.

- **Thesis:** *Misure di probabilità e spazi di Wasserstein* (in Italian, supervisor: A. Pisante)
- **Defense:** *viva* in Italian, October 29, 2013, with honors (*con lode*)

09/05–06/10 🎓 **High school diploma (*Diploma di Liceo Scientifico*)**, *Liceo Scientifico Statale Augusto Righi*, with honors (*con lode*).

Teaching experience

Lecturer, *Università degli studi di Roma "Tor Vergata"*, Rome, IT.

03/25–05/25 undergraduate lecture course, *Fisica, misura, errore e statistica*.

Lecturer, *Institute of Science and Technology Austria*, Klosterneuburg, AT.

04/24–06/24 graduate lecture course, *Optimal transport*.

04/22–06/22 graduate lecture course, *Fractional Gaussian Fields*, joint with F. Sau.

11/20–01/21 lecture course, *Maths for quantitative life scientists: Probability and Statistical Inference*.

Teaching assistant, *University of Bonn*, Bonn, DE.

04/19–07/19 undergraduate lecture course, *Algorithmische Mathematik II*, lecturer: Prof. Dr. P. Ferrari.

10/18–12/18 graduate lecture course, *Advanced Topics in Stochastic Analysis – Functional inequalities and stochastic analysis on manifolds*, lecturer: Prof. Dr. K. Kuwada.

04/18–09/18 graduate lecture course, *Stochastic Analysis*, lecturer: Prof. Dr. K. Kuwada.

10/16–03/17 graduate lecture course, *Introduction to Stochastic Analysis*, lecturer: Dr. M. Erbar.

Research papers

(1) L. Dello Schiavo and K. Suzuki. Persistence of Rademacher and Sobolev-to-Lipschitz Properties. *Adv. Math.*, 481:110542, 2025. 60 pp. doi:10.1016/j.aim.2025.110542

(2) L. Dello Schiavo, M. Magnabosco, and C. Rigoni. Gradient flows of (K, N)-convex functions with negative N. *Calc. Var. Part. Diff. Eq. (to appear, see arXiv:2412.04574)*, 2024. doi:10.48550/arXiv.2412.04574

(3) L. Dello Schiavo and V. Konarovskyi. Ill-posedness of the pure-noise Dean–Kawasaki equation. *Electron. Commun. Probab.*, 30:1–9, 2025. doi:10.1214/25-ECP702

(4) L. Dello Schiavo, R. Herry, E. Kopfer, and K.-T. Sturm. Polyharmonic Fields and Liouville Quantum Gravity Measures on Tori of Arbitrary Dimension: from Discrete to Continuous. *Math. Nachr.*, 298(1):244–281, 2025. doi:10.1002/mana.202400169

(5) L. Dello Schiavo, R. Herry, E. Kopfer, and K. Sturm. Conformally invariant random fields, Liouville quantum gravity measures, and random Paneitz operators on Riemannian manifolds of even dimension. *J. London Math. Soc.*, 110(5), October 2024. 80 pp. URL: http://dx.doi.org/10.1112/jlms.70003, doi:10.1112/jlms.70003

(6) L. Dello Schiavo, R. Herry, and K. Suzuki. Wasserstein geometry and Ricci curvature bounds for Poisson spaces. *J. École Poly.*, 11:957–1010, 2024. 55 pp. doi:10.5802/jep.270

(7) L. Dello Schiavo, J. Maas, and F. Pedrotti. Local Conditions for Global Convergence of Gradient Flows and Proximal Point Sequences in Metric Spaces. *Trans. Amer. Math. Soc.*, 377(6):3779–3804, 2024. 25 pp. doi:10.1090/tran/9156

(8) L. Dello Schiavo. A Characterization of Maps of Bounded Compression. *Math. Commun.*, 29:137–142, 2024. doi:10.48550/arXiv.2304.11348

(9) L. Dello Schiavo, E. Kopfer, and K.-T. Sturm. A Discovery Tour in Random Riemannian Geometry. *Potential Anal.*, 61:501–553, 2024. doi:10.1007/s11118-023-10118-0

(10) L. Dello Schiavo, L. Portinale, and F. Sau. Scaling Limits of Random Walks, Harmonic Profiles, and Stationary Non-Equilibrium States in Lipschitz Domains. *Ann. Appl. Probab.*, 34(2):1789–1845, 2024. doi:10.1214/23-AAP2007

(11) L. Dello Schiavo and E. W. Lytvynov. A Mecke-type Characterization of the Dirichlet–Ferguson Measure. *Electron. Commun. Probab.*, 28:1–12, 2023. doi:10.1214/23-ECP528

(12) L. Dello Schiavo and M. Wirth. Ergodic Decompositions of Dirichlet Forms under Order Isomorphisms. *J. Evol. Equ.*, 23(9), 2023. 22 pp. doi:10.1007/s00028-022-00859-7

(13) T. Cremaschi and L. Dello Schiavo. Effective Contraction of Skinning Maps. *Proc. Amer. Math. Soc. B*, 9:445–459, 2022. doi:10.1090/bproc/134

(14) L. Dello Schiavo. The Dirichlet–Ferguson Diffusion on the Space of Probability Measures over a Closed Riemannian Manifold. *Ann. Probab.*, 50(2):591–648, 2022. 58 pp. doi:10.1214/21-AOP1541

(15) L. Dello Schiavo and K. Suzuki. Sobolev-to-Lipschitz Property on QCD-spaces and Applications. *Math. Ann.*, 384:1815–1832, 2022. 18 pp. doi:10.1007/s00208-021-02331-2

(16) L. Dello Schiavo and K. Suzuki. Rademacher-type theorems and Sobolev-to-Lipschitz properties for strongly local Dirichlet spaces. *J. Funct. Anal.*, 281(11):109234, dec 2021. 63 pp. doi:10.1016/j.jfa.2021.109234

(17) L. Dello Schiavo. Ergodic Decomposition of Dirichlet Forms via Direct Integrals and Applications. *Potential Anal.*, 58:573–615, 2023. 43 pp. doi:10.1007/s11118-021-09951-y

(18) L. Dello Schiavo. A Rademacher-type theorem on L^2-Wasserstein spaces over closed Riemannian manifolds. *J. Funct. Anal.*, 278(6):108397, 2020. 51 pp. doi:10.1016/j.jfa.2019.108397

(19) L. Dello Schiavo. Characteristic functionals of Dirichlet measures. *Electron. J. Probab.*, 24(115):1–38, 2019. doi:10.1214/19-EJP371

(20) E. Agliari, A. Barra, L. Dello Schiavo, and A. Moro. Complete integrability of information processing by biochemical reactions. *Sci. Rep.*, 6:36314, 2016. doi:10.1038/srep36314

(21) L. Dello Schiavo. Marcinkiewicz continuity estimates for infinite energy solutions of some second order elliptic Dirichlet problems. *B. Unione Mat. Ital.*, 8:97–112, September 2015. 16 pp. doi:10.1007/s40574-015-0028-y

(22) E. Agliari, M. Altavilla, A. Barra, L. Dello Schiavo, and E. Katz. Notes on stochastic (bio)-logic gates: computing with allosteric cooperativity. *Sci. Rep.*, 5:9415, 2015. doi:10.1038/srep09415

Preprints

[23] L. Dello Schiavo and G. Brigati. Non-bilinear Dirichlet Functionals: Markovianity, locality, invariance. *arXiv:2510.02942*, 2025. doi:10.48550/arXiv.2510.02942

[24] L. Dello Schiavo and G. E. Sodini. The Hellinger–Kantorovich metric measure geometry on spaces of measures. *arXiv:2503.07802*, 2025. 96 pp. doi:10.48550/arXiv.2503.07802

[25] L. Dello Schiavo. Massive Particle Systems, Wasserstein Brownian Motions, and the Dean–Kawasaki Equation. *arXiv:2411.14936*, 2024. 103 pp. doi:10.48550/arXiv.2411.14936

[26] L. Dello Schiavo and F. Quattrocchi. Multivariate Dirichlet Moments and a Polychromatic Ewens Sampling Formula. *arXiv:2309.11292*, 2023. doi:10.48550/ARXIV.2309.11292

[27] L. Dello Schiavo and K. Suzuki. Configuration Spaces over Singular Spaces II – Curvature. *arXiv:2205.01379*, 2022

[28] L. Dello Schiavo and K. Suzuki. Configuration Spaces over Singular Spaces I – Dirichlet-Form and Metric Measure Geometry. *arXiv:2109.03192*, 2021. 85 pp.

Proceedings

{28} L. Dello Schiavo. A Discovery Tour in Random Riemannian Geometry. In J. Kerner, M. Täufer, and P. Yatsyna, editors, *Mini-Workshop: A Geometric Fairytale full of Spectral Gaps and Random Fruit*, volume 19 of *Oberwolfach Rep.*, pages 3051–3092. EMS Press, 2023. doi:10.4171/OWR/2022/53

{29} L. Dello Schiavo. Conformally Invariant Random Fields, Quantum Liouville Measures, and Random Paneitz Operators on Riemannian Manifolds of Even Dimension. In M. Gordina, T. Kumagai, L. Saloff-Coste, and K.-T. Sturm, editors, *Heat Kernels, Stochastic Processes and Functional Inequalities*, volume 19 of *Oberwolfach Rep.*, pages 2811–2893. EMS Press, 2023. doi:10.4171/OWR/2022/49

{30} L. Dello Schiavo. The Dirichlet–Ferguson Diffusion on the Space of Probability Measures over a Closed Riemannian Manifold. In M. Gordina, T. Kumagai, L. Saloff-Coste, and K.-T. Sturm, editors, *Heat Kernels, Stochastic Processes and Functional Inequalities*, volume 16 of *Oberwolfach Rep.*, pages 3348–3350. EMS Press, 2020. doi:10.4171/OWR/2019/54

Popularization of Science

{31} L. Dello Schiavo and A. E. Baccaglini-Frank. La quantità del nulla. *Ithaca*, X, 2017. In Italian

{32} G. Cecchetti and L. Dello Schiavo. Nature of language vs. language of nature – The limits of human languages in describing nature. In A. Altamore and G. Antonini, editors, Galileo and the Renaissance Scientific Discourse*, proceedings of the* First Roma Workshop on Past and Present Perceptions of Science, Rome, May 6, 2009. Ed. Nuova Cultura, 2010

Research stays

06/25 **CIRM (Trento)**, 1 week, FBK Research in Pairs programme with A. Pinamonti
02/25 **Technische Universität München**, 3 days, invitation by J. Zimmer
01/25 **Technische Universität Berlin**, 3 days, invitation by B. Gess and N. Perkowski
12/24 **Università Statale di Milano**, 2 days, invitation by S. Stuvard and F. Cavalletti
11/24 **Università di Trieste**, 1 week, invitation by F. Sau
07-08/24 **IAM Bonn**, 1 week, invitation by L. Portinale and M. Gordina
07/24 **IAM Bonn**, 3 days, invitation by L. Portinale
06/24 **Universität Hamburg**, 4 days, invitation by V.V. Konarovskyi
02/24 **Università di Roma 'Tor Vergata'**, 5 days, invitation by D. Marinucci
11/23 **Università di Trento**, 3 days, invitation by A. Pinamonti
11/23 **Universität Münster**, 4 days, invitation by M. Huesmann
11/23 **Keiō University**, 4 days, invitation by H. Kawabi
10/23 **Bielefeld University**, 4 days, invitation by M. Röckner
05/23 **Swansea University**, 5 days, invitation by E. W. Lytvynov
03/23 **IAM Bonn**, 1 week, invitation by L. Portinale
02/23 **GSSI**, 1 day, invitation by S. Cenatiempo and A. Nota
01/23 **Sapienza – Università di Roma**, 3 days, invitation by A. Faggionato
07/22 **Swansea University**, 5 days, invitation by E. W. Lytvynov
04/22 **IAM Bonn**, 1 week, invitation by K.-T. Sturm
11-12/21 **MFO**, 4 weeks, MFO Research in Pairs programme with K. Suzuki
11/21 **IAM Bonn**, 1 week, invitation by K.-T. Sturm
10/19 **Institute of Science and Technology Austria**, 2 weeks, invitation by J. Maas
05/19 **Sapienza – Università di Roma**, 5 days, invitation by L. Bertini & A. Faggionato
02/18 **Università degli Studi di Pavia**, 3 days, invitation by F. Bassetti
11/17 **Sapienza – Università di Roma**, 5 days, invitation by A. Faggionato
12/16 **Swansea University**, 5 days, invitation by E. W. Lytvynov

Invited talks

23-26/09/25	Workshop *Young Researchers in Stochastic Analysis and Stochastic Geometric Analysis*, EPFL, Lausanne
08/07/25	Mini-Workshop Optimal Transport for Complex Data, University of Vienna, Vienna
06/25	Conference **PRIN Geometry of RAndom FIelds and Applications (GRAFIA)**, Cortona
10/04/25	Seminario di probabilità, Sapienza – Università di Roma (part II)
02/04/25	Seminario di probabilità, Sapienza – Università di Roma (part I)
26/03/25	Seminar series **Random geometry and quantum gravity**, Institut Henri Poincaré (IHP), virtual
12/2/25	Stochastics Seminar, TU Munich, Munich
30/1/25	Stochastics Seminar, TU Berlin, Berlin
17/12/24	**Seminario di Analisi Matematica**, Università Statale di Milano, Milano
22/11/24	Workshop *Probability Event: Point Processes, Exchange & Research*, Münster
25/06/24	**Probability Kolloquium**, Hamburg
08/02/24	**RoMaDS Seminar**, Università 'Tor Vergata', Roma
20/12/23	Conference *Mean field interactions with singular kernels and their approximations*, IHP, Paris
28/11/23	**Seminario di Analisi**, Trento
22/11/23	**Oberseminar Mathematische Stochastik**, Münster
9/11/23	Conference *Stochastic Analysis*, RIMS, Kyōto
30/10/23	**Tōkyō Probability Seminar**, Tōkyō
12/10/23	**Bielefeld Stochastic Afternoon**, Bielefeld
22/09/23	Bielefeld University
8/09/23	XXII Congress of the Unione Matematica Italiana, Pisa
25/05/23	**Probability Seminar**, Swansea University
21/03/23	**Stochastic Analysis & Geometry Seminar**, IAM Bonn
20/02/23	**SMAQ – Seminario di Matematica L'Aquila**, GSSI
10/01/23	**Seminario di probabilità**, Sapienza – Università di Roma
1/12/22	MFO mini-workshop *A Geometric Fairytale full of Spectral Gaps and Random Fruit*, MFO
4/11/22	MFO workshop *Heat Kernels, Stochastic Processes and Functional Inequalities*, MFO
24/10/22	*Optimal Transportation and Applications*, Centro di Ricerca Matematica E. De Giorgi, Pisa
02/08/22	Three research days on "random geometries and interacting processes", HIM Bonn
28/07/22	**Probability Seminar**, Swansea University
16/06/22	Third Italian Meeting on Probability and Mathematical Statistics, Università di Bologna
21/04/22	**Stochastic Analysis & Geometry Seminar**, IAM Bonn
17/03/22	**SPDE Seminar**, TU Berlin, virtual
19/01/22	**PDE Afternoon**, University of Vienna, virtual
10/12/21	MFO mini-workshop *Variable Curvature Bounds, Analysis and Topology on Dirichlet Spaces*
21/09/21	**Seminario di probabilità**, Sapienza – Università di Roma
9/02/21	**Pisa Online Seminar**, virtual
12/01/21	**Vienna Probability Seminar**, virtual
21/11/19	MFO workshop *Heat Kernels, Stochastic Processes and Functional Inequalities*, MFO
31/10/19	Institute of Science and Technology Austria
05/09/19	JAP-GER Conference on Stochastic Analysis 2019, Fukuoka University
06/05/19	**Seminario di probabilità**, Sapienza – Università di Roma
11/03/19	Collaborative Research Center 1060 Retreat 2019, Project B03
28/02/18	Università degli Studi di Pavia
28/11/17	**Seminario di probabilità**, Sapienza – Università di Roma

01/12/16 **Probability Seminar**, Swansea University

Service

Organizer.

7–9/4/25 Workshop *Stochastic equations and particle systems*, INdAM, Roma, IT

21–24/5/24 Workshop *Synthetic Curvature Bounds for Non-Smooth Spaces: Beyond Finite Dimension*, ESI, Vienna, AT

17–21/7/23 SPP2265 Summer School *Probability and Geometry on Configuration Spaces*, Harnack-Haus, Berlin, DE

Grants and funding information

11/26 **Royal Society International Exchanges 2025 Global Round 2 — Travel Grant**, joint with Dr. Federico Cornalba, University of Bath, UK.

- Project: *SPDEs for Fluctuating Hydrodynamics: Efficient simulation of multi-scale multi-species particle systems*
- Funding: £11.620,00

07/22–10/24 **FWF ESPRIT Fellow**, *Institute of Science and Technology Austria*, Klosterneuburg, AT.

- Project: *Configuration Spaces over Non-Smooth Spaces*
- Funding: € 294.015,98

07/22 **APART-MINT Fellow** of the Austrian Academy of Sciences **(declined)**.

- Project: *Multi-dimensional Wasserstein Diffusions*
- Funding: € 70.000,00

Memberships, honors, and awards

2023–present Member of the *Unione Matematica Italiana* (UMI)

2020–present Member of the *Schwerpunktprogramm* (*Priority Programme*) 2265 *Random Geometric Systems* of the Deutsche Forschungsgemeinschaft (German Research Foundation)

2018–2019 participation in Bonn Graduate Center's *Doctorate Plus* Training Program
Access to a series of soft skills training workshops.

2016–2018 Member of the Italian *National Research Group in Mathematical Physics* of the *National Institute of Advanced Mathematics* (GNFM – INdAM)

2014–2015 M.Sc. Honor classes (*Percorso d'Eccellenza*), Sapienza – Università di Roma.
Access to one PhD course, reserved to the most proficient students (about 5% of the total), with a final supervised dissertation.
Dissertation: *Takesaki's proof of a Tannaka–Stinespring–Tatsuuma-type Pre-duality for Commutative Measured Involutive Hopf–von Neumann Algebras* (in English, supervisor: C. Pinzari)

2011–2013 B.Sc. Honor classes (*Percorso d'Eccellenza*), Sapienza – Università di Roma.
Access to supplementary courses on advanced topics reserved to the most proficient students (about 10% of the total), each with a final test to pass.

Other Projects

WDP Wolfram Demonstration Project

- *Zonotope Construction via Shephard's Theorem*
- *Homogeneous Splines*
- *Asymmetric Simple Exclusion Processes* (with A. Occelli)

Refereeing activity

2016–present ∘ *Scientific Reports* ∘ *Electronic Journal of Probability* ∘ *Journal of Functional Analysis* ∘ *Electronic Communications in Probability* ∘ *Demonstratio Mathematica* ∘ *Probability Theory and Related Fields* ∘ *Manuscripta Mathematica* ∘ *Stochastic and Partial Differential Equations* ∘ *Stochastic Processes and Applications* ∘ *Quarterly Journal of Mathematics*

Selected attended schools, conferences, seminars, talks

19–22/11/24	Workshop *Probability Event: Point Processes, Exchange & Research*, Münster, DE
18–22/12/23	Conference *Mean field interactions with singular kernels and their approximations*, IHP, Paris, FR
6–10/11/23	Conference *Stochastic Analysis*, RIMS, Kyōto, JP
23–26/10/23	21st Symposium *Stochastic Analysis on Large Scale Interacting Systems*, RIMS, Kyōto, JP
4–9/09/23	XXII Congress of the Unione Matematica Italiana, Pisa, IT
24–28/7/23	*Stochastic Processes and Applications*, Lisbon, PT
27–30/3/23	SPP 2265 *Random Geometric Structures* Second Annual Conference, DLR, Cologne, DE
27/11–3/12/22	MFO 2248b *A Geometric Fairytale full of Spectral Gaps & Random Fruit*, Oberwolfach, DE
30/10–5/11/19	MFO 2244 *Heat Kernels, Stochastic Processes and Functional Ineq.s*, Oberwolfach, DE
24–28/10/22	10th conference on *Optimal Transportation and Applications*, CRM E. De Giorgi, Pisa, IT
7–9/9/22	Austrian Stochastic Days, Vienna, AT
29/8–3/7/22	Joint conferences: New Trends in Dirichlet Forms and Optimal Transport & From Dirichlet Forms to Wasserstein Geometry, HCM, Bonn, DE
1–3/8/22	Three research days on "random geometries and interacting processes", HIM, Bonn, DE
13–16/6/22	Third Italian Meeting on Probability and Mathematical Statistics, Bologna, IT
11–14/4/22	SPP 2265 *Random Geometric Structures* First Annual Conference, Harnack-Haus, Berlin, DE
7–12/3/21	BIRS 5184 *Stochastics and Geometry (Online)*, Banff, CA (virtual)
17–23/11/19	MFO 1947 *Heat Kernels, Stochastic Processes and Functional Ineq.s*, Oberwolfach, DE
2–6/7/19	9th Japanese-German Open Conference on *Stochastic Analysis*, Fukuoka, JP
7–12/7/19	5th School on *Stochastic and Geometric Analysis*, Piz Buin, AT
11–13/3/19	Retreat 2019 of the Collaborative Research Center 1060, Weilburg, DE
12–16/11/18	9th conference on *Optimal Transportation and Applications*, SNS, Pisa, IT
12–14/3/18	Retreat 2018 of the Collaborative Research Center 1060, Bad Neuenahr-Ahrweiler, DE
4–29/9/17	*Metric Measure Spaces and Ricci Curvature* - Intense Activity Period, MPIM, Bonn, DE
7–11/11/16	8th conference on *Optimal Transportation and Applications*, SNS, Pisa, IT
10–14/10/16	*Stochastic Partial Differential Equations and Related Fields*, Bielefeld U, Bielefeld, DE
29/8–2/9/16	Follow-up Workshop to JTP *Optimal Transportation*, HIM, Bonn, DE
4–8/04/16	4th School on *Stochastic and Geometric Analysis*, Vlieland, NE
21–23/10/15	*Panorama of Mathematics*, HCM, Bonn, DE

Languages

Italian (native), English (proficient), German (proficient)